

Q₁

[red, blue, blue, red, green, grey]

a. red blue green grey

red	1	0	0	0
blue	0	1	0	0
blue	0	1	0	0
red	1	0	0	0
green	0	0	1	0
grey	0	0	0	1

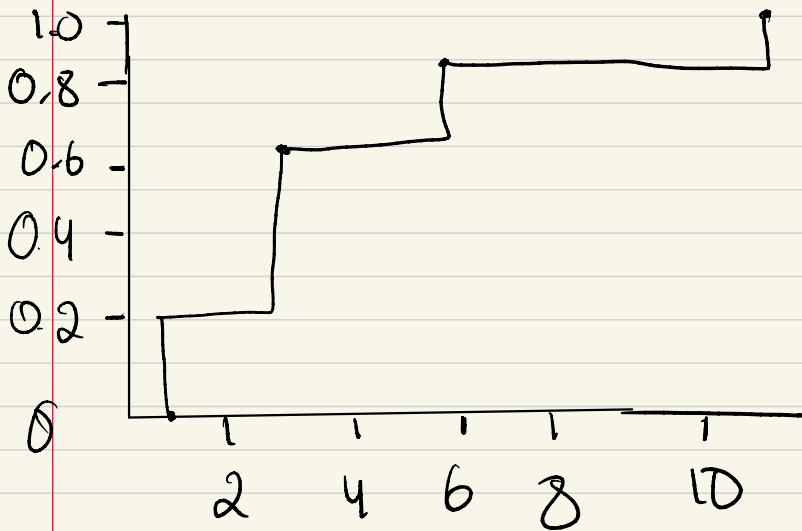
b. red = $\frac{2}{6}$

blue = $\frac{2}{6}$

green = $\frac{1}{6}$

grey = $\frac{1}{6}$

$$C. X = [1, 3, 3, 6, 11]$$



$$\text{For Jump} = 1/n = 1/5 = 0.2$$

$$1 = \text{jump of } 1/5 = 0.2$$

$$3 = \text{jump of } 2/5 = 0.2 + 0.4 = 0.6$$

$$6 = \text{jump of } 3/5 = 0.6 + 0.2 = 0.8$$

$$11 = \text{jump of } 4/5 = 0.8 + 0.2 = 1.0$$

Q2

random variable $X \Rightarrow Y = a + bX$
with $b > 0$

Distribution of $X \Rightarrow F_X(x)$

Distribution of Y , $F_Y(y)$.

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(a + bX \leq y) \\ &= P(bX \leq y - a) \\ &= P\left(X \leq \frac{y - a}{b}\right) \end{aligned}$$

$$F_Y(y) = F_X\left(\frac{y - a}{b}\right)$$

Density

$$\begin{aligned} f_Y(y) &= f_X\left(\frac{y-a}{b}\right) \\ &= \left(f_X\left(\frac{y-a}{b}\right)\right) \left[\frac{y-a}{b}\right]' \\ &= f_X\left(\frac{y-a}{b}\right) \left(\frac{1}{b}\right) \\ &= \frac{1}{b} f_X\left(\frac{y-a}{b}\right) \end{aligned}$$

b. $X = [1, 3, 4]$

$$X = \frac{1+3+4}{3} = \frac{8}{3} \approx 2.667.$$

$$\begin{aligned} c. \quad X^2 &= (1)^2 + (3)^2 + (4)^2 = 1 + 9 + 16 \\ &= \underline{26/3} \end{aligned}$$

$$\sqrt{\quad} = 8.667$$

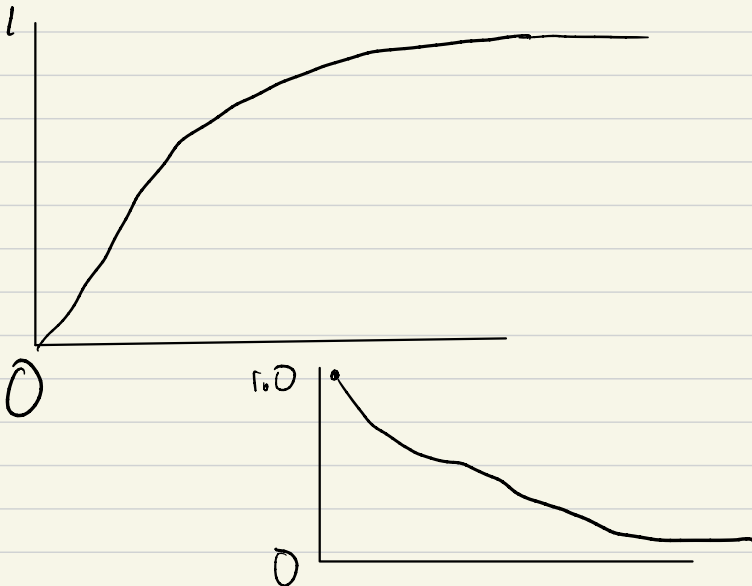
$$d. \chi^2 = (2.067)^2 = 7.11$$

This mean does not equal to the mean of the square from part c.

Q_3

$$F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-x} & x \geq 0 \end{cases}$$

a_0



$$b. P(X \geq 3)$$

$$P(X \geq 3) + P(X < 3) = 1$$

$$P(X \geq 3) = 1 - P(X < 3)$$

$$P(X \geq 3) = 1 - f(3)$$

$$= 1 - (1 - e^{-3})$$

$$= 1 - 0.950$$

$$\approx 0.05$$

$$P(X \leq 35) = F(35)$$

$$= 1 - e^{-35}$$

$$= 1$$

$$c. \quad Y = 1 + 3X$$

$$F_Y(y) = P[Y \leq y]$$

$$= P[1 + 3X \leq y]$$

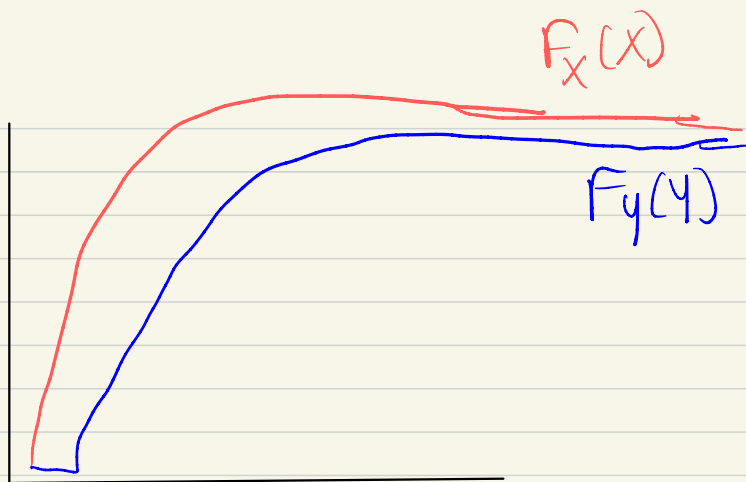
$$= P[3X \leq y - 1]$$

$$= P\left[X \leq \frac{y-1}{3}\right]$$

$$= P_X\left[\frac{y-1}{3}\right]$$

$$f_Y(y) = f_X\left(\frac{y-1}{3}\right)$$

$$= 1 - e^{-\left(\frac{y-1}{3}\right)}$$



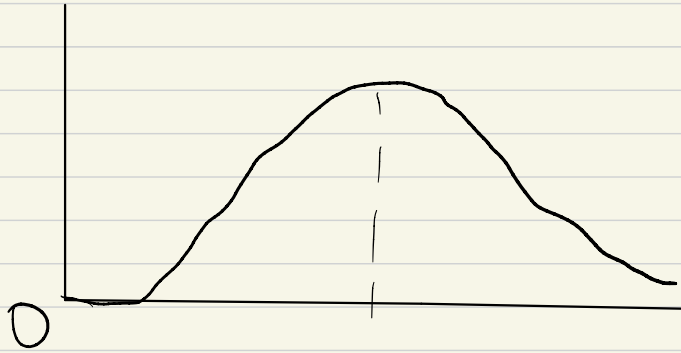
$$P(Y < 10) = 1 - e^{-\left(\frac{10-1}{3}\right)} = 1 - e^{-\frac{9}{3}} \\ = \underline{\underline{0.950}}$$

Q4

$$f(x) = \frac{1}{1 + e^{-ax}}$$

a. Logistic distribution





$$\begin{aligned} \text{b. } P(X \geq 0.8) &= 1 - P(X < 0.8) \\ &= 1 - F(0.8) \\ &= 1 - \frac{1}{1 + e^{-x}} \\ &= 1 - \frac{1}{1 + e^{-0.8}} \\ &= 0.310 \end{aligned}$$

$$\begin{aligned} P(X \leq 0.3) &= F(0.3) \\ &= \frac{1}{1 + e^{-0.3}} \approx 0.574 \end{aligned}$$

$$c. \quad Y = 2X - 1$$

$$F_Y(y) = P[Y \leq y]$$

$$= P[2X - 1 \leq y]$$

$$= P[2X \leq y + 1]$$

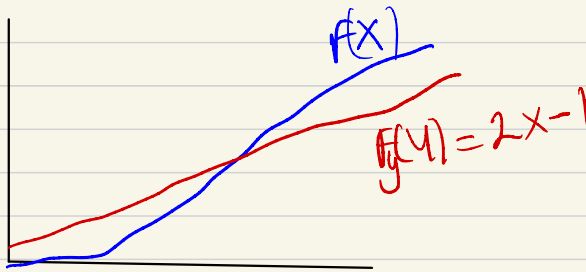
$$= P\left[X \leq \frac{y+1}{2}\right]$$

$$= F_X\left(\frac{y+1}{2}\right)$$

$$F_Y(y) = F_X\left(\frac{y+1}{2}\right)$$

$$= \frac{1}{1 + e^{-\left(\frac{y+1}{2}\right)}}$$

$$F_Y(y) = P[Y \leq y] = \frac{1}{1 + e^{-\left(\frac{y+1}{2}\right)}}$$



$$\begin{aligned} P(Y < 0) &= f(0) \\ &= \frac{1}{1 + e^{-\left(\frac{0+1}{2}\right)}} \\ &= \frac{1}{1 + e^{-1/2}} \\ &= 0.622 \end{aligned}$$

Q5

$$a. \quad 1 - e^{-x} = \frac{1}{2}$$

$$e^{-x} = \frac{1}{2} - 1$$

$$e^{-x} = -\frac{1}{2}$$

$$\ln(e^{-x}) = \ln\left(\frac{1}{2}\right)$$

$$-x = \ln\left(\frac{1}{2}\right)$$

$$x = -\ln\left(\frac{1}{2}\right)$$

$$x \approx 0.693$$

Median of exponential distrib^o ≈ 0.693

$$b. \quad \frac{1}{1+e^{-x}} = \frac{1}{2}$$

$$1+e^{-x} = 2$$

$$e^{-x} = 2-1$$

$$e^{-x} = 1$$

$$\ln(e^{-x}) = \ln(1)$$

$$x = -\ln(1)$$

$$x = 0$$

Median of logistic distribution is 0.

$$c. \quad 1 - e^{-x} = 0$$

$$e^{-x} = -(0-1)$$

$$e^{-x} = -0+1$$

$$\ln(e^{-x}) = \ln(1-u)$$

$$-x = \ln(1-u)$$

$$x = -\ln(1-u)$$

$F^{-1}(u) = -\ln(1-u)$
↳ Quantile function of exponential distribution.

$$\text{d. } \frac{1}{1+e^{-x}} = u$$

$$u(1+e^{-x}) = 1$$

$$1+e^{-x} = \frac{1}{u}$$

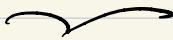
$$e^{-x} = \frac{1}{u} - 1$$

$$\ln(e^{-\infty}) = \ln\left(\frac{1}{0} - 1\right)$$

$$-\infty = \ln\left(\frac{1}{0} - 1\right)$$

$$\infty = -\ln\left(\frac{1}{0} - 1\right)$$

Q6



[https://github.com/
NieciaS/
uu_categorical_and_num
eric/blob/main/
problem.ipynb](https://github.com/NieciaS/uu_categorical_and_numeric/blob/main/problem.ipynb)